



MODULE 6 UNIT 2

Activity submission scenario

UNIVERSITY OF CAPE TOWN
BUSINESS SYSTEMS ANALYSIS



Instructions

Although this scenario document expands on the scenarios of Modules 1 and 3, the information in those modules is contextual. Only the information in this scenario document is necessary for the activity submission in this module.

Note:

This scenario is entirely fictitious. Although many details are inspired by real life, resemblance to any real-life person or entity or event is purely coincidental and unintentional.

Scenario

Now in your sixth week at Kabeljou, the project to implement the Sell It solution, with its e-commerce and CRM modules, is well underway. The business case you developed, with the clear, factual feasibility of Sell It over WebSiteShopify, helped create a sense of certainty for Kabeljou's top management.

This, in turn, has developed a strong, nearly unanimous commitment for the solution, which you not only noticed in the way your colleagues arrive on time for your meetings, but also in the way that they seem much more prepared and engaged than in the first few weeks of the project – even Miriam has made surprisingly positive comments about the new system!

Speed bumps in the SDLC

So far, one of the challenges in your various workshops on requirements discovery and validation is that you still spend a lot of time trying to get everyone on the same page before the “real” work of each workshop starts. Your colleagues often get distracted from meetings' agendas by re-clarifying business processes. This is because some of your workshops are with people who are infrequently involved in the project, like the finance team or the customer service representatives (CSRs).

The two primary causes of this inefficiency are, firstly, that there's a lot of detail to know. You reflect that maybe you've been expecting a bit too much of people who aren't involved with the processes as intimately as you are. Secondly, the processes themselves change with each workshop as you continuously validate and re-design improvements to them.

To add to your worries, many processes are still quite foreign to you, especially because each business process seems to have so many different ways of flowing, depending on who is telling you about it. It seems as if you're the only person in the organisation who is uncertain, while everyone else seems to be at ease with their own singular perspectives. Perhaps it's like bugs on a beach ball – each bug sitting on its own stripe sees the world with full certainty as only blue, or red, or yellow, while you see all the different colours.



Your insight into the many conflicting processes and requirements is causing you some anxiety about exactly how to specify the processes for the developer to configure in the system.

From all of this, it is obvious to you that you need to be able to make a quick reference to key business processes in your workshops. There are no old process models on which to base your new designs, so each model needs to be created afresh. This means you have complete freedom in not only mapping out each process, but in establishing the general standards for how process models are diagrammed at Kabeljou. The BPMN standard is the most logical starting point.

Although you feel uneasy about the ambiguities and complexities in this space of “the great unknown”, as is typical at the start of many SDLCs, you reassure yourself that it is entirely normal and comes with the BSA territory. To help calm your feelings of uncertainty, you bolster your confidence by reminding yourself that no one else at Kabeljou has your training, knowledge and tools to solve the problems you’re discovering – you’re the best person for this job!

Sales process

The business process you’ve prioritised to be modelled first is a high-level overview of the sales process, because it is so often needed in your workshops and interviews. You intend using this model in conversations with end users and the project’s sponsor, Marius. You’ve collated details acquired from Marius, end users and other sources, including job descriptions and the detailed financial policies that Miriam has drafted.

In a meeting with Marius in Anton’s office – remember, despite being MD, Marius has no office of his own – he sketched a flowchart of the sales process overview on Anton’s flip chart. You’ve stuck Marius’ flowchart on your office wall and, for ease of reference in your documentation, you took a photo of it, as shown in Figure 1.

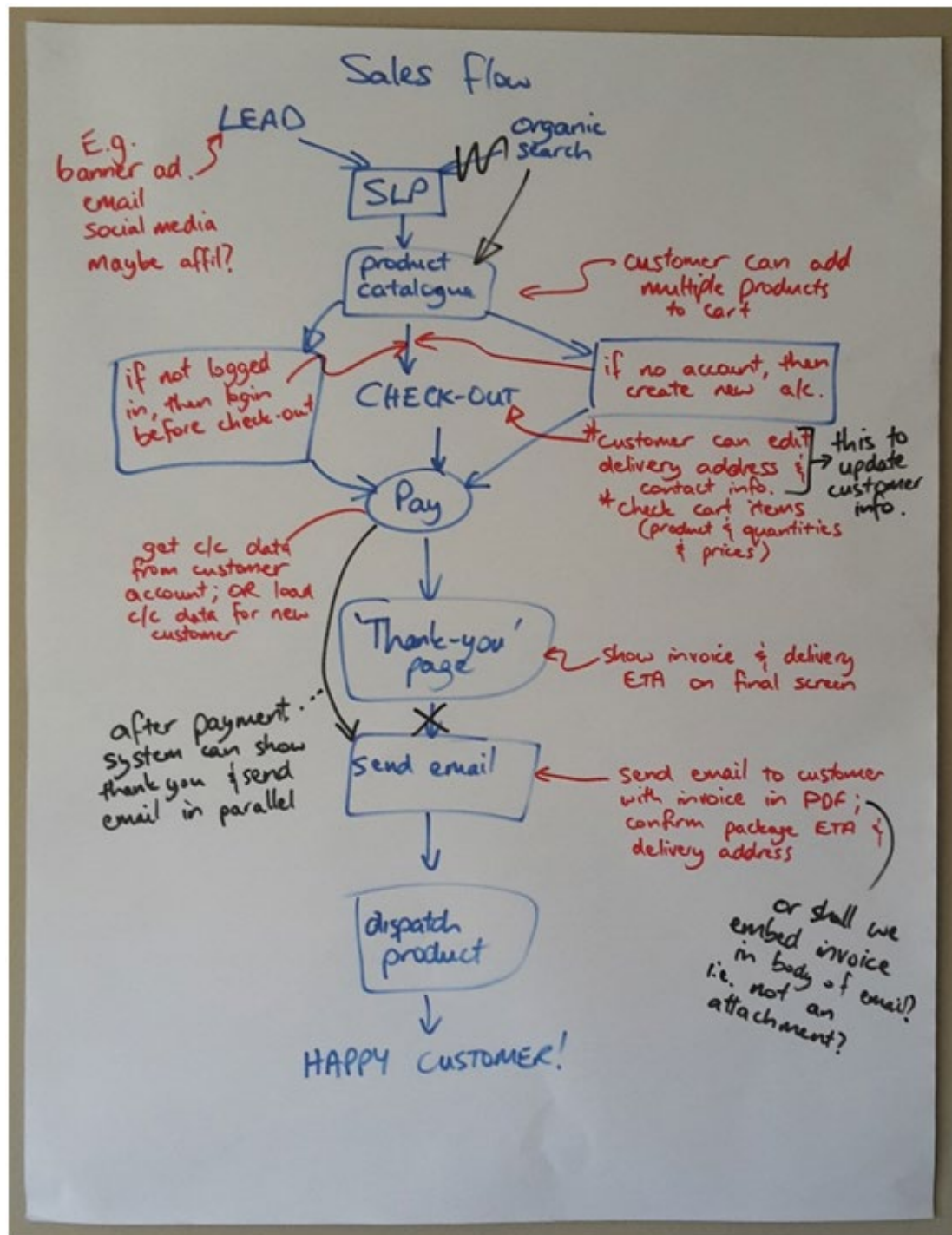


Figure 1: Marius' sales flow chart.

As part of your diligent approach, you documented the key points of Marius' interview, which were partially based on your rough hand-written notes that you took during the interview, plus what you could add after the meeting.



Here are those key points:

1. The sales process discussed with Marius is not the as-is process, but the process to be implemented in Sell It.
2. Customers can enter the sales process in one of two ways: via a link or without a link.
3. Marius's use of "LEAD" as one of the start conditions to the sales process is confusing because *all* entry points into the sales process are to be considered leads in the sales funnel. Agreed with Marius to rather use the term "link" for linked leads.
4. A link can be from any lead generation activity that we create, for example, an online banner ad, Facebook ad, Google ad, a link sent in an email campaign, or a link in a Facebook post.
5. Each link must have the marketing campaign code and, when we start affiliate marketing, also an affiliate code. Together with other data collected from each user's visit – like user device and software data, session data, user IP address and so on – this data is collectively known as tracking data and will be used in marketing analytics and financial controls, such as fraud audits and paying affiliate marketers for sales to customers they refer to us.
6. The system must pass the tracking data through each step in the process and be included in the order record that is created when payment is successful so that management can track, report on and analyse marketing strategies.
7. A visitor who enters the sales process using a link will enter the e-commerce site (and sales process) by being taken to a sales landing page (SLP). The SLP is what the link points to.
8. All non-linked leads into the sales funnel will be categorised as "Browse". E.g., a visitor who enters the catalogue through an organic search, such as a search engine query, or through any offline advertising like an ad in print publications like *Stywe Lyne / Tight Lines*, or by referral from an existing customer.
9. A customer who enters the sales process from browsing will enter the product catalogue pages directly and not come in via an SLP.
10. On the catalogue pages, the visitor can add any product(s) to the cart, even if navigating through many different products' pages.
11. At any time while in the catalogue, the visitor can hit the "Check out now" button. When the customer clicks "Check out now", the system must check if the customer is logged in or not.
12. If the visitor is not logged in, and if the visitor does not yet have a login account, then force a new customer record to be created and automatically log the customer in after



creating the new account. Only the customer's email address, password and first name are required at this stage when creating a new account. The email address will be the customer's username to login.

13. The system flow can take one of three paths from the catalogue to check out: 1) already logged in, so go directly to check out; 2) log in using an existing customer account; 3) create a new customer account and auto login.
14. In the checkout page, new customers (who created their account in order to reach the checkout page) must add their last name, delivery address and other contact information in order to proceed to the payment page. Information for existing customers must be auto-populated in the checkout page using the data already stored in their customer record.
15. Any new or changed information that a customer enters in the checkout screen must automatically update their customer record, whether for a new or an existing customer.
16. In the checkout page, a customer can also remove a product entirely or change the quantity of each item ordered. The billing total, excluding delivery charges, must be updated on the screen after each change.
17. When the customer is satisfied with the cart, they can click "Pay now" to proceed to the credit card payment page.
18. Sell It Online is PCI DSS-compliant, so credit card data can be stored by Sell It. "PCI DSS" stands for "Payment Card Industry's Data Security Standard". Without DSS compliance, credit card data would not be allowed to persist beyond the transaction, i.e., it could not be stored in a database and the system must "forget" the data immediately after the transaction.
19. Similar to the checkout page, the payment page must auto-fill the fields with the last-used credit card data for existing customers. New customers must enter their card data. Again, if there is new card data, including changed data for existing customers, the customer's record must be updated with this.
20. Once the customer confirms payment information, in other words, when the customer clicks the "Pay now" button, we must get the authorisation code from the bank and, if we get it, immediately collect the payment. Once payment is received, simultaneously present the "Thank you" page and send a confirmatory email to the customer. The "Thank you" page and the email must both show the invoice with products, quantities and billing amounts, the delivery address, and the package's expected time of arrival (ETA).
21. We have not yet decided if the email will contain the invoice as a PDF attachment or if the invoice should be embedded as text directly in the email's body.

22. Up until we get the credit card authorisation code, and the payment step is complete, a customer can exit the sales process at any point. E.g., not creating a new account after clicking the “Check out” button in the catalogue, or not clicking “Pay now” in the checkout page will mean that the payment process does not continue. If the cart contains at least one product and the process is not completed, even for a new customer with no login account, then this must be considered an abandoned cart. Marius wants to be able to analyse the contents of abandoned carts and where in the process customers abandon their carts.
23. The sales process ends successfully after the “Thank you” page is shown, and the email is sent.
24. Because we will not be implementing Sell It’s shipping module yet, after completing the transaction, the system must trigger an email to Kadys’ team in dispatch to notify them of the order that must be picked, packed and shipped.

A simple, uncluttered process model

In reviewing Marius’ flow chart together with your detailed notes, it is obvious that you have too much information for a sales process overview model. In fact, you have a lot of detail that belongs on a modelling layer below the layer of a sales overview process model. In a follow-up conversation with Marius, you outlined and validated what you feel are the key activities and conditions in the overview model. Marius has confirmed the following:

25. Show tasks for only these steps:
 - 25.1. Present the sales landing page i.e., any product catalogue page, whether the visitor enters as a Link or a Browse user.
 - 25.2. Manage the cart.
 - 25.3. When the customer clicks “Check out now”, check whether the customer is logged in already and, if not, either force the login or force creating a new account and then automatically log in to proceed to check out.
 - 25.4. Present the cart and user data on screen.
 - 25.5. Handle the payment.
 - 25.6. Present the “Thank you” page and also send the email to the customer.
26. Because almost no one else at Kabeljou has been exposed to BPMN and process modelling, you have shrewdly chosen to use the simplest possible notation in the BPMN 2.0 set of elements. This means that you will prepare your sales overview process model using only these versions of BPMN 2.0’s elements:



- 26.1. “None” start and end conditions. For example, no intermediate conditions, conditional or message conditions.
- 26.2. Only simple tasks with no markers or decorations in them. That is, no sub-processes, transactions and call activities; no loop markers; no decorations like “service task” or “user task”, and so on.
- 26.3. Only exclusive, inclusive and parallel gateways. For example, no event-based or complex gateways, and so on.
- 26.4. Omit data elements and annotations; the “cleaner” the model’s visual appearance, the better.
- 26.5. No swim lanes; only a pool for the sales process.
- 26.6. Show meta-data for document control, model identification and traceability.

At some point after creating this simple, clean view of the sales process overview, you might create more detailed versions of the same process or parts of the process, such as the payment handling process (described in the next section). Whatever detail that belongs in lower level models will be restricted to those lower level models, which you plan to develop later, separately from the overview process model.

Your task now is to use Marius’ flowchart and your requirements analysis notes to prepare a sales overview process model using “simple” BPMN.

A technical overview of the system

You’ve now presented the key business processes to stakeholders in various workshops and your work has been well worth the effort – the project is now moving forward with everyone’s full commitment. This is a relief because, before this, a few colleagues had expressed their doubts about their value. You attribute this sceptical disposition to your colleagues being confident that each of their views of the business processes was the *only* view and they were not aware that others had different views of the same processes.

Although you’ve managed to get everyone on the same page regarding each key business process, there is still some confusion amongst your colleagues about how everything inside the system hangs together and relates to things outside the system, like users and integration with external systems. Realising that a unified vision of the overall system is important in ensuring everyone collaborates effectively, you commit to developing a systems architecture model (“SAM”) that shows the software elements, or modules, inside the Sell It system.

Before drawing the SAM, you reviewed all your consulting notes to make sure you have recorded every important detail about the modules and processes and that they are as accurate and as up to date as possible.



Here's what you could extract from your notes:

27. The main modules (AKA components) inside the "Sell It E-Commerce System" are the modules for:

- 27.1. **Customer;**
- 27.2. **Catalogue;**
- 27.3. **Order;**
- 27.4. **Campaign;**
- 27.5. **Reporting;**
- 27.6. **Payment;** and
- 27.7. **Shipping** (but the "Ship It" module won't be implemented yet, so it's out of scope for this project).

28. There are only two external systems to which integration will be required:

- 28.1. Each bank **gateway**: this is the mechanism that will be used to collect customers' credit card payments.
- 28.2. The **email** system: this will be used to email order confirmations to customers. Also, because the shipping module is not being implemented, emails will be sent to our dispatch team in the warehouse as a means of notifying them when an order is received and its products are to be picked, packed and dispatched.

29. We've identified seven different roles for human actors that will interact with the system:

- 29.1. **Customers** will need to create a customer account in the Customer module. Each order a customer places will be handled in the Order module.
- 29.2. **CSRs** need access to the Customer and Order modules so that they can provide customer support.
- 29.3. **Management** will need to access various reports using the Reporting module.
- 29.4. **Dispatch** must be able to access each order in the Order module so that they can update it after dispatching an order's shipment.
- 29.5. **Marketing** clerks will work with two modules: they will manage stock items and marketing campaigns in the Catalogue and Campaign modules respectively.



- 29.6. The **finance** clerks will use the Payment module to manage the payment gateway data for each bank.
- 29.7. Lastly, the **system administrator** role will access the whole system to maintain it.

Use this summary of your consulting notes to create your SAM for the whole solution that is in scope for this project. For your convenience, a starter SAM is shown in Figure 2. It is not accurate in that what is shown here does not match the specifications needed in your SAM, but you can use this design format for the elements of the system, components and entities. Or you can design your own elements. Whichever design you choose, be sure to show the central system, its components and the relevant external entities that the system connects to.



Figure 2: System architecture model (incomplete and inaccurate) for Sell It at Kabeljou.